



The
BRITISH
UNIVERSITY
IN EGYPT

26CSSE03I

Game Design and Development

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GitHub Lab

PART 1: Setup and configuration

How to set up Git and GitHub?

Git is essentially a remote memory card where you are able to upload your files and save your progress as you work (version control).

GitHub is a website where you can upload all your "saved" work so others can see it and collaborate with you as well as it being a way of saving it via the cloud.

Steps:

1. download git onto pcs using <https://git-scm.com/downloads/win>
2. run through the installer normally leaving everything at default.
3. open <https://github.com> and create an account

Git Bash is a CLI that allows interaction with Git using Linux commands. Once you download Git you will need to configure it so that you are able to access your GitHub account through your pc locally and confirm the email that is associated with your commits.

How to configure Git bash?

Steps:

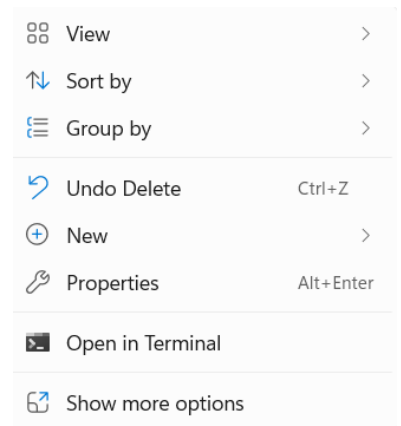
Right click, choose open in Terminal.

Type in the following command individually. Don't expect any outcome as this is just configuration without conformation.

Configuration commands→

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your.email@example.com"
```



PART 2: How to create a project on GitHub?

A project on GitHub is called a Repository.

Repository is essentially a centralized storage location that will carry all the files and details of a project on the cloud so anyone can access it.

Steps:

1. Log into GitHub account.
2. Click on New to create a new repository.
3. Type in repository details (name and description), choose visibility option and create.

Notes:

- Public repositories are visible to anyone and can be found by searching

Owner name/Repository name

- Private repositories are only visible to owners and collaborators.
- For other people to be able to add files and collaborate with you on a project, they will have to be added in the collaborators section.

(settings in repository → collaborators → add people)

- Adding a README file is optional however it is best practice to do so.

---Watch Video 1 for Details on previous steps---

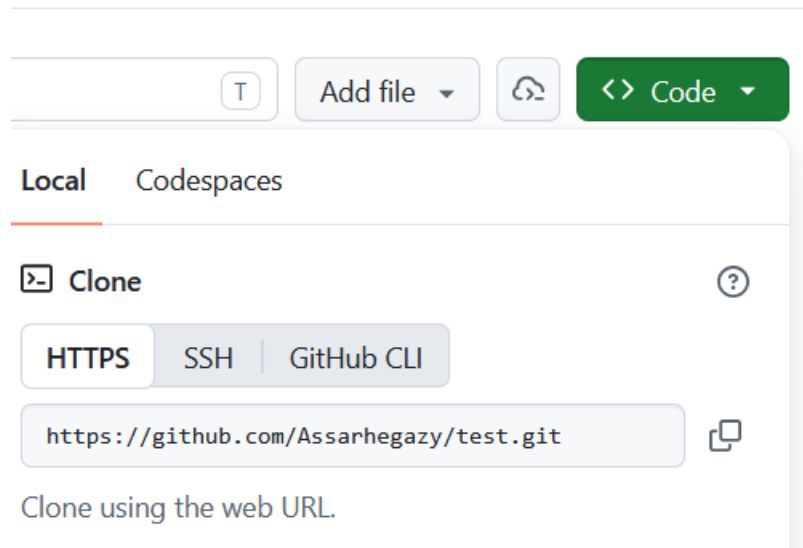
PART 3: How to access an already existing project from GitHub locally on your PC?

To get access to a repository that exists on GitHub you will need to **Clone** it onto your pc.

Clone is copying a repository onto your local pc with access to add and modify files online.

Steps:

1. Log into GitHub account.
2. Search for the repository required. → You will only find the publicly available ones.
3. Click on **Code** and copy the https URL to clone.
4. Open desktop or any file you want your project to be in, right click, open in Terminal
5. Type in this command → **git clone (add the copied URL here)**
To paste the URL use **SHIFT + INSERT** keys.
6. A folder with the name of the repository is cloned.



Step 3

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\assar\OneDrive\Desktop\test> git clone https://github.com/Assarhegazy/test.git
Cloning into 'test'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\assar\OneDrive\Desktop\test> |
```

Step 5

PART 4: How to create and work on Branches?

To keep all the work organized and avoid mishaps with code/file timelines, branches are used.

Branch is an isolated line of development where you can bring in a new feature, test and experiment without touching or messing with the main code base which should be stable at all times. It allows for multiple developers to work in parallel and helps with version control.

Steps:

1. Log into GitHub account.
2. Open the main page of your repository.

3. Find the branches tab and create a new branch.

---Watch Video 2 for Details on previous steps---

Always switch to your branch before doing any work.

To ensure that you are on your branch, you can check in vs code the branch name.

---Watch Video 3 for Details on previous steps---

PART 5: How to commit and push work on Branches?

After finishing all the work changes, you will need to commit and push these changes to your branch to save them.

Commit is saving all your changes on your local repository on your PC.

Push is updating the remote repository with whatever was committed on your local one, allowing others to see it too.

Steps:

1. Open vs code, select repository folder.
2. Open source control tab to view changes.
3. Stage the changes that are ready to be committed using the plus sign.
4. Write a commit message and then click on commit and push. **(always check you are committing to your own branch)**

You will be able to view your commit on the repository's webpage on GitHub.

---Watch Video 4 for Details on previous steps---

PART 6: How to create a pull request and merge changes?

After committing to your own branch, you will have to merge these changes with the main branch that has the stable code. To merge your branch with the main branch, you will create a pull request from the repository webpage on GitHub.

Pull request essentially is asking the owner of the repository to pull and review the changes you made and committed to your branch onto the main branch to join the stable code.

Steps:

1. Open repository webpage on GitHub
2. Open your own branch and view your newest commit.
3. Create a pull request and write message.

Once the pull request is made, it must be merged and reviewed so long as there are no conflicts. The owner of the repo will review these changes and merge the pull request indicating that whatever changes present on your branch are now available on the main branch too.

---Watch Video 5 for Details on previous steps---

Note: There is a possibility that your branch doesn't match the main branch for different reasons such as:

1. Modifications to the same file and part have been made e.g. 2 different people work on the same scene changing things or writing the same script.
2. Someone pushed changes before your own thus the main branch has data that you haven't pulled just yet.

In that case a **merge conflict** is created in which GitHub is unable to automatically merge files. Merge conflicts must be resolved manually by developers in which they view the changes that created a conflict to be resolved either on GitHub or the editor and choose how to deal with them.

PART 7: How to pull from a Branch?

To ensure that your branch is up to date with what is on the main branch you will need to pull from the main.

Pull is getting the newest changes from the remote repository to your local one.

Steps:

1. Open vs code, select repository folder.
2. Open source control tab.
3. Click on 3 dots to the side of your repository file in the tab.
4. Choose the option pull from
5. Choose the main branch to get the latest updates
6. Sync up the changes

---Watch Video 6 for Details on previous steps---

PART 8: GitHub tips

1. Always pull from the main before working on your branch
2. Never work on the same file with someone else
3. .gitignore files are used when there are file that don't need to be tracked and shouldn't be present on your repository.

E.g.: files such as library, build etc. are computer/editor specific files created by unity unique to each user, these files can create a conflict and shouldn't be added to your repo. To avoid pushing them, a .gitignore file is added to your project specifying which folders should be skipped during a commit and push.

To add a .gitignore file:

1. Open repository webpage on GitHub
2. Add a new file and type in the name field .gitignore
3. Choose a template

